

ATPL EASA STUDY GUIDE | 050 - METEOROLOGY

Chapter 14: Meteorological Reports & Forecasts (METAR, TAF, SIGMET)

Syllabus Source: CAE Oxford (pp. 473-510) | Pass Target: >= 90% AviationExam

1. METAR & SPECI DECODING RULES

METAR is an aerodrome routine meteorological report issued every 30 or 60 minutes. SPECI is a special report issued when weather deteriorates across critical thresholds:

- Wind Group: e.g. 24015G28KT = Mean wind 240° True at 15 kt, gusting to 28 kt. 180V260 = Wind direction varying between 180° and 260°.
- Visibility & RVR: e.g. 1400 = Prevailing visibility 1,400 meters. R27/0800VP1500U = Runway 27 RVR varying from 800 m to greater than 1,500 m, with upward trend (U).
- Present Weather Codes: Descriptors: TS (Thunderstorm), FZ (Freezing), SH (Shower); Precipitation: DZ (Drizzle), RA (Rain), SN (Snow), GR (Hail >= 5 mm); Obscuration: FG (Fog), BR (Mist), HZ (Haze).
- Cloud Coverage & Base: FEW = 1-2 octas, SCT = 3-4 octas, BKN = 5-7 octas (CEILING!), OVC = 8 octas (CEILING!). Base given in hundreds of feet AGL (e.g. BKN015 = Broken at 1,500 ft AGL). Only CB and TCU cloud types are appended.
- CAVOK Conditions (All 4 must be satisfied!): 1. Visibility >= 10 km; 2. No cloud below 5,000 ft or Minimum Sector Altitude (MSA); 3. No CB or TCU; 4. No significant weather phenomena!

2. TAF (TERMINAL AERODROME FORECAST)

- Validity Periods: Short TAF: 9 hours (issued every 3h). Long TAF: 24 to 30 hours (issued every 6h).
- Change Groups: BECMG: Gradual change over the specified time window. TEMPO: Temporary fluctuations lasting < 1 hour and covering < 50% of the period. FM (From): Rapid permanent change at the exact hour and minute.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): METAR / TAF Rapid Operational

Example: LEMD 121430Z 24015G28KT 180V280 4000 R32L/1200D +TSRA FEW010 BKN025CB 18/14 Q1008 BECMG 6000 NSW= Step 1: Station & Time: Madrid Barajas, 12th at 14:30 UTC. Step 2: Wind: 240 true at 15 kt, gusts to 28 kt, varying between 180 and 280. Step 3: Visibility: 4,000 meters general; RVR on runway 32L is 1,200 m decreasing (D). Step 4: Weather: Heavy thunderstorm with rain (+TSRA). Step 5: Clouds: FEW at 1,000 ft, Broken ceiling at 2,500 ft Cumulonimbus (BKN025CB). Step 6: Temp/Dewpoint: 18 C / 14 C; QNH 1008 hPa. Step 7: Trend: Becoming 6,000 m, No Significant Weather (NSW).

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Complete METAR & TAF Flight Dispatch Decoding

SCENARIO (EASA Flight Deck Dispatch Drill): A flight crew reviews the destination

weather: METAR LEMD 121430Z 04018G32KT 010V080 3000 +SHRA SCT008 BKN018CB 14/11 Q1008

NOSIG= QUESTION: Decode all items in this METAR step-by-step for flight

dispatch. SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Header -> LEMD (Madrid Barajas), 12th day of the month at 14:30 UTC. Step 2: Wind -> 04018G32KT = Mean wind 040° True at 18 knots, gusting to 32 knots! - 010V080 = Wind direction is varying clockwise/counter-clockwise between 010° and 080°. Step 3: Visibility -> 3000 = Minimum prevailing visibility is 3,000 meters. Step 4: Weather -> +SHRA = HEAVY (+) SHOWERS (SH) OF RAIN (RA)! Step 5: Clouds: - SCT008 = Scattered (3 to 4 octas) at 800 ft AGL (NOT a ceiling). - BKN018CB = Broken (5 to 7 octas) at 1,800 ft AGL containing CUMULONIMBUS (CB)! This constitutes an official CEILING at 1,800 ft with severe convective hazards! Step 6: Temperatures -> 14/11 = Outside Air Temperature +14°C, Dew Point +11°C (narrow 3°C spread). Step 7: Pressure -> Q1008 = QNH is 1008 hPa. Step 8: Trend -> NOSIG = No significant change expected during the next 2 hours. FINAL ANSWER: Madrid has gusty winds to 32 kt, heavy rain showers, visibility 3 km, and CB cloud ceiling at 1,800 ft AGL.

[!] EXAM TRAP (AVIATIONEXAM) - The Ceiling Definition Trap

- By official ICAO definition, a 'CEILING' is the height above ground of the lowest layer of clouds covering MORE THAN HALF THE SKY: - BKN (5-7 octas) = CEILING. - OVC (8 octas) = CEILING. - FEW (1-2 octas) and SCT (3-4 octas) are NOT ceilings!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Which of the following conditions MUST be met simultaneously for 'CAVOK' to be reported in a METAR?

- [A]: Visibility 5,000 m and no low clouds
- [B]: Visibility ≥ 10 km, NO CLOUD below 5,000 ft or MSA (whichever is higher), NO CB or TCU, and NO significant weather
- [C]: Clear sky and zero wind
- [D]: Zero fog

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. CAVOK (Ceiling And Visibility OK) requires ≥ 10 km visibility, absence of clouds below 5,000 ft / MSA, no CB/TCU at any level, and no significant weather phenomena.

Q2: In aviation meteorology, which cloud coverage codes constitute an official 'CEILING'?

- [A]: Only OVC (Overcast)
- [B]: BROKEN (BKN: 5 to 7 octas) and OVERCAST (OVC: 8 octas)
- [C]: SCT and BKN
- [D]: FEW and SCT

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under ICAO Annex 3, a cloud ceiling is defined as the height of the base of the lowest layer covering more than 4 octas (i.e. BKN or OVC).

Q3: In a TAF forecast, what does the change indicator 'TEMPO 1216 2000' signify?

- [A]: Permanent change to 2,000 m visibility from 12:00 to 16:00
- [B]: TEMPORARY fluctuations to 2,000 meters visibility occurring between 12:00 UTC and 16:00 UTC, each lasting less than 1 hour and covering in aggregate less than half the period
- [C]: Temperature will reach 20°C
- [D]: Wind 200° at 16 kt

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. TEMPO denotes temporary weather fluctuations lasting less than 60 minutes per instance and less than 50% of the total time window.